

Pre-test Placement Exam 2026

Name: _____ email: _____

Start time: _____ End time: _____

Program (circle one): MACSS QMSA MAPSS OTHER

Please show all your work and write as clearly / legibly as possible! **GOOD LUCK!!!!**

1 Basics / Fundamentals

1.1 (1 point) Evaluate:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

1.2 (2 points) Concavity:

Sketch the general shape of the following function on $(-4,0)$, including any max/min values, and whether/where the function is concave up / down. $f(x) = 2x^3 + 3x^2 - 12x$. Label all critical values for the domain of $(-4,0)$.

2 Calculus: integrals and derivatives

2.1 (1 point) Evaluate $\int (16y^3 + 8y)(y^4 + y^2 + 3)^3 dy$

2.2 (1 point) Evaluate $\int 3xe^{3x} dx$.

3 Linear algebra

3.1 Find the Jacobian and Hessian of the following: $k(x, y) = (4 + xy)(x + y) + 2$

a. (1 point) Jacobian

b. (1 point) Hessian

3.2 (1 point) Interpret what the Hessian and Jacobian tell us in the context of a function $f(x, y)$. Explain in 1-2 sentences.

3.3 (2 points) Find the determinant and eigenvalue(s) of the following matrix:

$$\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$$

a. (1 point) Determinant:

b. (1 point) Eigenvalue(s):

3.4 (1 point) Find the determinant of the following matrix:

$$\begin{pmatrix} 5 & 0 & 3 \\ 3 & 3 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

3.5 Suppose we have the following: $\mathbf{A}\mathbf{Y}\mathbf{B} = \mathbf{C}$ and we know that $\mathbf{C}$ exists.

- a. (1 point) If $\mathbf{A}$ has two rows and $\mathbf{B}$ has the dimensions of 3×2 , what is the resulting dimension of $\mathbf{C}$? If it is not knowable, explain why.

- b. (1 point) Using the matrix information from above, if you wanted to isolate $\mathbf{Y}$, how would you do so? Provide an explanation and resulting equation, including any assumptions you would need to make. If it is not possible, explain why.

4.3 (2 points) Find the value of b that makes this a valid pdf. Provide two reasons for why / how you know it is a pdf. If it's not possible, state why.

$$\int_0^1 \int_0^b (6x^2y) dy dx$$

4.4 Marginal Probability: for the function $f(x, y) = 8x + 8y$ find the following when both x and y fall within $[0, 0.5]$:

a. (0.5 points) $f_X(x)$

b. (0.5 points) $f_{Y|X}(y|x)$

c. (1 point) Are $f_X(x)$ and $f_Y(y)$ independent? How do you know if they are or are not?

4.5 (2 points) Suppose you recently took a flu test and received a negative result. Let's assume that the false negative rate for flu tests is 0.4. Furthermore, suppose approximately 1 in 8 people have the flu. Because the test was created specifically to detect the flu, the probability of a false positive is very small, around 0.01.

Given this information, what is the chance that you have the flu, despite a negative test? Be sure to show all your work.

4.6 Suppose the following:

Function	Value
$E[D]$	5
$E[F]$	4
$E[DF]$	10
$\text{Var}(D)$	16
$\text{Var}(F)$	25

a. (0.5 points) What is $\text{Cov}(D, F)$?

b. (0.5 points) What is the correlation between (D) and (F) ?

5 Statistics

Z statistics of interest

z^*	$1 - \alpha$	$\alpha/2$	α
± 1.65	90%	0.05	0.10
± 1.96	95%	0.025	0.05
± 2.58	99%	0.005	0.01

5.1 (2 points) What is the sampling distribution of the mean and how is it different from the sample distribution?

Specifically state the mean and variance of each and how they differ, if at all. Be sure to include WHY they are or are not different. Provide approximately 4 sentences in your explanation.

5.2 You're exploring the connection between streaming new music and subscription service users, surveying 50 total users. You find that among 25 YouTube users, the average is 15.04 albums/month with a standard deviation of 2.52. Among 25 Apple Music streamers, the average is 16.42 albums/month with a standard deviation of 2.34.

- (1 point) Provide a point estimate of the difference
- (2 points) Is this difference statistically meaningful? Show all your work/steps, any assumptions you made, and explain your reasoning in approx 3 sentences.
- (1 point) Provide a 90% confidence interval using your point estimate from above.

5.3 Scratch paper area

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